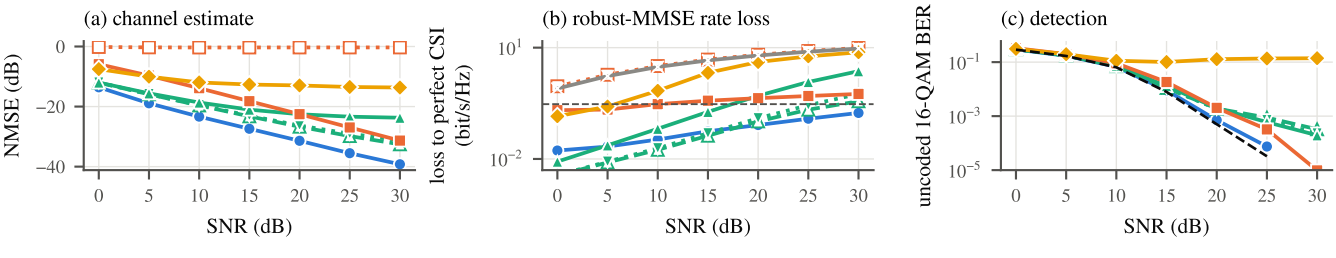
(a) channel estimate
NMSE (dB)
0
−20
−40
0
5
10
15
20
25
30
SNR (dB)
(b) robust-MMSE rate loss
loss to perfect CSI (bit/s/Hz)
10^{1}
10^{-2}
0
5
10
15
20
25
30
SNR (dB)
(c) detection
uncoded 16-QAM BER
10^{-1}
10^{-3}
10^{-5}
0
5
10
15
20
25
30
SNR (dB)
Proposed (timing only)
Exact-element LMMSE
Element model 20 MHz off
Ideal-element LMMSE
CE-BEM, all h, $N_p = 32$
CE-BEM, $|h| \leq 128$
OMP
One-tap
Perfect CSI